

Kyrylo Batrak

Software Engineer (C++/Python)

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EDUCATION

Taras Shevchenko National University of Kyiv (KNU)

Kyiv, Ukraine

Bachelor of Science in Computer Mathematics (Mechanics and Mathematics Faculty)

Sep. 2024 – Present

- **Relevant Coursework:** Algorithms & Data Structures, C/C++ Programming, Python Programming (including OOP), Concrete Mathematics, Mathematical Analysis, Multidimensional Analysis, Analytical Geometry, Discrete Mathematics, Linear Algebra, Algebra & Number Theory, Combinatorial Analysis, Probability Theory, Differential Equations, Mathematical Logic, Finite Fields (Galois Fields).

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, GNU Octave.
- **Computer Science:** Data Structures, Algorithms, OOP, Software Architecture.
- **Mathematics:** Linear Algebra, Discrete Mathematics, Combinatorics, Number Theory, Finite Fields (Galois Fields), Mathematical Analysis, Abstract Algebra, Analytical Geometry, Probability Theory, Differential Equations, Mathematical Logic.
- **Tools & Platforms:** Git/GitHub, CMake, L^AT_EX, Linux.
- **Languages:** Ukrainian (Native), English (B2).
- **Soft Skills:** Extensive experience in teamwork, strong leadership, analytical problem-solving, and effective communication.

SOFTWARE PROJECTS

Library Management System (C++) — *OOP, CMake, STL, File Serialization*

- Engineered a console-based application with a clean architecture and custom CSV serialization for persistent data storage.
- Utilized CMake for cross-platform builds and modern C++17 STL for dynamic inventory tracking and role-based user management.
- Integrated the `<ctime>` library to automate transaction histories and accurately calculate overdue fines.

ArrayStatistic: C/C++ Statistical Analysis Library — *Memory Management, Algorithms, Make*

- Developed a parallel statistical library implementing complex mathematical tests (Spearman, Fisher, Z-test) from scratch.
- Demonstrated deep understanding of low-level memory management in C and modern RAII/STL in C++.
- Designed a comprehensive cross-testing suite using Makefiles to guarantee logical consistency between modules.

Pygame Tetris: Classic Arcade Game (Python) — *Python, Pygame, MVC Architecture*

- Designed a fully functional game featuring matrix collision detection, piece rotation algorithms, and dynamic line clearing.
- Structured the project using MVC-inspired architecture to cleanly separate core logic, UI rendering, and event handling.
- Built a custom UI interface and integrated a robust SoundManager class for asynchronous background music.

COMPETITIONS

Student Programming Olympiad named after S.O. Lebedev and V.M. Glushkov (KPI-OPEN)

Kyiv, Ukraine

Participant

Jul. 2026

- Competed in solving complex algorithmic challenges and optimizing code performance under strict time limits.
- Applied advanced knowledge of data structures (such as trees and graphs) and algorithmic theory to develop efficient solutions.

CERTIFICATIONS

Grade Education Centre

General English B2 Upper Intermediate: Course 1

General English B1+ Intermediate

Kyiv, Ukraine

May 2026

2025 – 2026

Cisco Networking Academy

C++ Essentials 1 & 2; Python Essentials 1 & 2

Kyiv, Ukraine

2024 – 2025

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Secretary General of the Student Parliament

Mechanics and Mathematics Faculty, KNU

2026 – Present

Kyiv, Ukraine

- Coordinate the administrative and organizational operations of the Student Parliament, managing official documentation and meetings.
- Facilitate effective communication between student representatives, faculty administration, and university committees to support student initiatives.

Head of the Electoral Commission

Mechanics and Mathematics Faculty, KNU

2024 – 2026

Kyiv, Ukraine

- Led and managed the electoral processes at the faculty level, ensuring absolute transparency, fairness, and strict adherence to university regulations.
- Demonstrated high-level teamwork and organizational skills by coordinating a dedicated committee, resolving operational challenges, and facilitating smooth communication with both students and university administration.